L6 MF Revision Paper 1

Q1.

Find arg $(-4 - 7i)$ to the nearest degree.

Circle your answer.

$-120°$ $\qquad$ $-60°$ $\qquad$ $30°$ $\qquad$ $60°$

(Total 1 mark)

Q2.

$z = 3 - \mathrm{i}$

Determine the value of zz^*

Circle your answer.

10 $\qquad$ $\sqrt{10}$ $\qquad$ $10 - 2\mathrm{i}$ $\qquad$ $10 + 2\mathrm{i}$

(Total 1 mark)

Q3.

Which of the following matrices is singular?

Circle your answer.

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\qquad$ $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ $\qquad$ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $\qquad$ $\begin{bmatrix} 1 & -2 \\ 1 & 2 \end{bmatrix}$

(Total 1 mark)

Q4.

Point P has polar coordinates $\left(2, \frac{2\pi}{3}\right)$.

Which of the following are the Cartesian coordinates of P?

Circle your answer.

$(1, -\sqrt{3})$ $\qquad$ $(-\sqrt{3}, 1)$ $\qquad$ $(\sqrt{3}, -1)$ $\qquad$ $(-1, \sqrt{3})$

(Total 1 mark)

Q5.

A reflection is represented by the matrix $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

State the equation of the line of invariant points.

Circle your answer.

$x = 0$ $\qquad$ $y = 0$ $\qquad$ $y = x$ $\qquad$ $y = -x$

(Total 1 mark)

Q6.

Solve the equation $2z - 5iz^* = 12$

(Total 4 marks)

Q7.

A and **B** are non-singular square matrices.

(a) Write down the product $\mathbf{AA}^{-1}$ as a single matrix.

(1)

(b) **M** is a matrix such that **M** = **AB** .

Prove that $\mathbf{M}^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$

(3)

(Total 4 marks)

Q8.

The matrices **A** and **B** are such that

$$\mathbf{A} = \begin{bmatrix} 2 & a & 3 \\ 0 & -2 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 & -3 \\ -2 & 4a \\ 0 & 5 \end{bmatrix}$$

(a) Find the product **AB** in terms of a.

(2)

(b) Find the determinant of **AB** in terms of a.

(1)

(c) Show that **AB** is singular when $a = -1$

(2)

(Total 5 marks)

Q9.

The matrices **A** and **B** are defined by

$$\mathbf{A} = \begin{bmatrix} p & 2 \\ 4 & p \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$$

(a) Find, in terms of p, the matrices:

(i) **A** − **B**;

(1)

(ii) **AB**.

(2)

(b) Show that there is a value of p for which **A** − **B** + **AB** = $k\mathbf{I}$, where k is an integer and $\mathbf{I}$ is the 2 × 2 identity matrix, and state the corresponding value of k.

(4)

(Total 7 marks)

Q10.

The line L has polar equation

$$r = \frac{k}{\sin\theta}$$

where k is a positive constant.

(a) Sketch L.

(1)

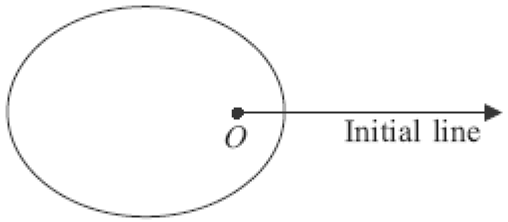
(b) State the minimum distance between L and the point O.

(1)

(Total 2 marks)

Q11.

The diagram shows a sketch of a curve C, the pole O and the initial line.



The curve C has polar equation

$$r = \frac{2}{3 + 2\cos\theta}, \quad 0 \leq \theta \leq 2\pi$$

(a) Verify that the point L with polar coordinates $(2, \pi)$ lies on C.

(1)

(b) The circle with polar equation $r = 1$ intersects C at the points M and N.

(i) Find the polar coordinates of M and N.

(3)

(ii) Find the area of triangle LMN.

(4)

(c) Find a cartesian equation of C, giving your answer in the form $9y^2 = \mathrm{f}(x)$.

(5)

(Total 13 marks)

Q12.

Find the sum of all the integers from 1 to 999 inclusive that are not square or cube numbers.

(Total 5 marks)

Q13.

(a) Use the standard formulae for $\sum_{r=1}^{n} r$ and $\sum_{r=1}^{n} r^2$ to show that

$$\sum_{r=1}^{n} r(r+3) = an(n+1)(n+b)$$

where a and b are constants to be determined.

(4)

(b) Hence, or otherwise, find a fully factorised expression for

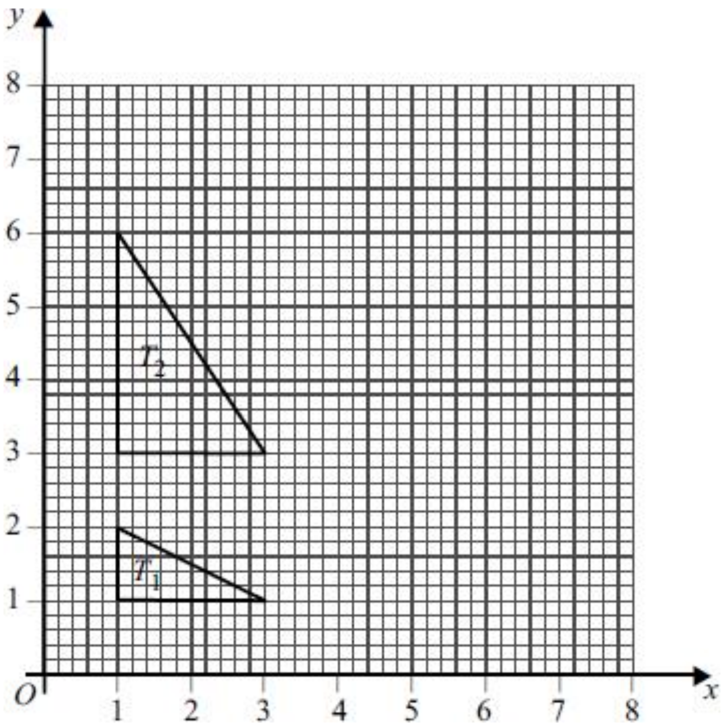
$$\sum_{r=n+1}^{5n} r(r+3)$$

(3)

(Total 7 marks)

Q14.

The diagram shows two triangles, T_1 and T_2.



(a) Find the matrix which represents the stretch that maps triangle T_1 onto triangle T_2.

(2)

(b) The triangle T_2 is reflected in the line $y = \sqrt{3}x$ to give a third triangle, T_3. Find, using surd forms where appropriate:

(i) the matrix which represents the reflection that maps triangle T_2 onto triangle T_3;

(2)

(ii) the matrix which represents the combined transformation that maps triangle T_1 onto triangle T_3.

(2)

(Total 6 marks)

Q15.

The matrix $\mathbf{C} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, where a and b are positive real numbers, and

$$\mathbf{C}^2 = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

Use **C** to show that $\cos\frac{\pi}{12}$ can be written in the form $\frac{\sqrt{\sqrt{m}+n}}{2}$, where m and n are integers.

(Total 7 marks)